Endogenous Innovation in the Tech Cold War Era

Tao Louie Xu[a,b] and Li Liu[b]

[a] *School of Social and Political Science, The University of Edinburgh, 15a George Square, Edinburgh, EH8 9LD, UK*

[b] *Fudan Development Institute, Fudan University, 220 Handan Road, Shanghai, 200433, China*

This review revisits endogenous innovation as a co-evolutionary process of dynamic capabilities amidst the global techno-nationalist Tech Cold War. Integrating micro organisational and macro institutional perspectives, it articulates a triple-staged capability growth chain: technology exploration, market exploitation, and ecosystem orchestration to capture the innovation-oriented adaptive ambidexterity by which emerging tech firms transition from component supplier to supply-chain ecosystem orchestrator. Proposing a research agenda of innovation ecosystem, mega-science infrastructure, regional cluster, and organisational culture, this research offers an alternative analytical approach to endogenous innovation in fractured environments across firm, regional, and national dimensions.

Keywords: endogenous innovation; Tech Cold War; neo-techno-nationalism; dynamic capabilities; industrial development; emerging-markets firms

1. What is Endogenous Innovation?

Endogenous innovation is an innovation paradigm materialised through a national or regional system with internally cultivated capabilities. Contrary to exogenous innovation characterised by global knowledge transfer and regeneration (Rakas & Hain, 2019), endogenous innovation manifests as the orchestration of available resources and capabilities to yield self-reliant innovation endeavour. This concept is rooted in the endogenous growth theory which formulates macroeconomic growth with domestically mobilised knowledge, human capital, and innovation investments rather than international trade or foreign technology transfer (Aghion & Howitt, 1992; Romer, 1994; Grossman & Helpman, 1994). This theoretical lineage situates innovation as an outcome of purposeful knowledge production process of *creative destruction* whereby new technologies supersede those obsolete.

Notwithstanding an explicit lineage, the wordage and scope conditions of endogenous innovation is often blurred and interchangeable with *indigenous innovation*. For our Tech War context, indigenous innovation is a sovereign act not contingent on foreign resources but strategic autonomy (Zhang et al., 2023), which our research regards as a geopolitical narrative alternative to endogenous innovation. However, although indigenous innovation implies a techno-nationalist overtone, endogenous innovation more accurately uncovers its essence—an alignment process across non-globally available resources, capabilities, and institutions (Edsand, 2019; Yu & Yan, 2021). To be specific, our endogeneity never connotes an absolutely insular internality confined in a single firm, region, or nation; it is the capacity of a system to autonomously internalise and reconfigure locally and regionally available resources into innovation whether with global linkages or not (Antonelli, 1998, 2000; Rakas & Hain, 2019). The *national innovation systems* conceptualise this actor-network alignment across universities, research institutes, governments, firms, and other organisations, where the innovation capacity is systemically cultivated within national or regional boundaries (Freeman, 1995; Furman et al., 2002). Within the innovation actor-networks, university, industry, and government actors are co-evolving (Etzkowitz & Leydesdorff, 2000); thus, endogenous innovation is not merely internal to firms but emerging from the *second frame* of innovation policy (Schot & Steinmueller, 2018)—the system and systematic interactions that reinforce systemic knowledge production. Our research interprets this interactive alignment as a substitute to the linear input-output paradigm of technology and market. As Antonelli (2017) notes, it is a *creative response* that proactively adapts and innovates in response to external shocks and constraints.

Organisational and collective capabilities are the starting junction to endogenous innovation. Collective capabilities of innovation systems, such as absorptive capacity, endogenously facilitate innovation without reliance on external linkages (Cooke et al., 1997). Firms embedded in such systems can exhibit adaptability by leveraging endogenous capabilities devoid of imported technologies, as China's state subsidies enhance innovation, catalysing internal processes of learning and changing (Howell, 2017). Similarly, when exogenous access is restricted due to export controls on advanced components, firms are inclined to recombine and exploit internal resources (Edsand, 2019; Zhang et al., 2023).Although open markets and international collaboration are important, open innovation never automatically erodes endogenous innovation systems but reinforces systematic innovation act if managed within a coherent systemic framework (Wang et al., 2012; Oh et al., 2016). When a system coordinates the selective absorption and adaptation of external knowledge, open innovation and free markets complement endogenous innovation. Erzurumlu et al. (2022) exemplify that the configuration of domestic systems of regulation, governance, and infrastructure interpret divergent innovation outcomes across economies. This interpretation reaffirms that endogenous innovation is not a unitary variable but a function of systemic coherence.

The concept of endogenous innovation offers a conceptual anchor for interpreting innovation as a locally, systemically orchestrated process of structural transformation. Particularly in times of techno-nationalist ruptures, grasping the means states mobilise endogenous innovation systems is essential. Yet, to articulate system's operations, it is vital to dissect both the micro-organisational foundations that enable firms to sense, seize, and reconfigure resources—and the macro-institutional environments that channel positive externalities throughout the system. Our following sections first examine the ambidextrous dynamic capabilities; then, the externalities of regional clusters and cross-sectoral networks. The following reveals the interplay of internal capability-building and outside-in alignment, reinforcing each other in the adaptive change of endogenous innovation.

1.1 Micro-foundation of Endogenous Innovation: Dynamic Capabilities and Organisational Ambidexterity

A tech firm's ambidextrous dynamic capabilities constitute its micro-foundation of endogenous innovation. *Dynamic capabilities*, a firm's abilities to effectively sense and seize opportunities by resource reconfiguration (Teece et al., 1997; Dodourova & Zhao, 2023), coexist and co-evolve with *organisational ambidexterity* of exploration and exploitation (Teece, 2007; O'Reilly & Tushman, 2008; Pundziene et al., 2022). Ambidexterity is simultaneously an outcome of, and conduit for, dynamic capabilities (Úbeda-García et al., 2020; Farzaneh et al., 2022). This co-evolutionary capability-building integrates sensing, seizing, and reconfiguring into adaptive exploration and exploitation, which is the first step towards endogenous innovation (Edsand, 2019).

From a micro-foundational perspective, endogenous innovation is not a trivial, necessary function of institutional environments, but fundamentally, together with firm growth, rooted in the dynamic capabilities and process of firms to accumulate, manage, and reconfigure internal resources in response to technological and market shifts. Confronted with a shift to techno-nationalist fragmentation, innovation actors cannot merely rely on passive technology transfer or exogenous spillovers; instead, cultivating the self-reliant innovation competency by sensing changes, seizing opportunities, and transforming asset configurations with agility is vital (Teece, 2007, 2022; Zhang et al., 2023). Further, the transition from imitation to internal capability-building reflects that innovation is no longer a linear trajectory of technology importation, modification, and subsequent catching-up (Reddy, 1997; Edgerton, 2007); such exogenous innovation is increasingly insufficient due to the fragmentation of global supply chains. In response, emerging-markets firms increasingly shift innovation endeavour towards endogenous capability-building (Liao & Li, 2018; Venugopal et al., 2019; Audretsch & Belitski, 2022). It implies, rather than supplicating on disrupted global linkages, enhancing internal resource utilisation, regional collaboration, and ecosystem engagement (Marra et al., 2019; Hwang et al., 2021; Simba et al., 2023; Xu & Hu, 2024). The enhanced capabilities not only cement technological sovereignty but sustain competitive advantage to navigate fragmented global value chains.

The capabilities of sensing is the firm capacity to (re)probe opportunities and threats, involving market research of customer demands, technological changes, and the structural reordering of the world economy (Teece, 2007; Petricevic & Teece, 2019). Dynamic sensing involves external *scanning* and outside-in *searching*. Externally, firms scan technological fronts, market demands, competitor actions, regulatory regimes, and the very significant external shocks triggered by geopolitical trends of techno-nationalism (Wilden et al., 2016; Luo, 2022a, 2022b; Zhang et al., 2023). In our case, it initiates firms to detect not only the challenges incurred by techno-nationalist value chain ruptures but transformative opportunities driven by economic multi-polarisation and emerging regionalism when the role of multinational knowledge transfer is redefined from globalisation to regionalised technological sovereign.

Following environmental scanning, searching enables firms to translate sensed opportunities into targeted endeavour to pinpoint and configure resources. It involves identifying underutilised knowledge assets and overcoming organisational silos to activate innovation capabilities (Teece, 2007; Petricevic & Teece, 2019); for another, it captures accessible, non-global yet strategically proximate resources since considerable technical know-how and market insight is embedded in regional open innovation networks—high-end producer services in advanced manufacturing clusters—which, though external in physics, function as available extensions of one's own ecosystem (Lassen & Laugen, 2017; Liao et al., 2020; Xu & Hu, 2024). Tech firms thus increasingly engage with regional innovation infrastructures as embedded actors leveraging dynamic capabilities to absorb and align these resources to internal goals (Marra et al., 2019; Audretsch & Belitski, 2022; Bettiol et al., 2023). Searching is thus an outside-in capability of process: catalysed by sensing yet realised through the internalisation of available resources—bridging internal recombination with place-based innovation access. In short, this two-way sensing substitutes weakened global linkages, embedding innovation in micro-firm perception and recombination for adaptive attention and strategic realignment.

Seizing is the next capability to translate sensed opportunities into concrete strategic actions and convert innovation capabilities into gains (Teece, 2007). It is the reallocation of organisational attention and resources to overcome structural inertia and invest in scalable solutions (Giniuniene & Pundziene, 2020; Lovallo et al., 2020). The ambidexterity of seizing refers to the simultaneous pursuit of exploratory innovation and the exploitation of available assets. Tech firms navigate the tension between R&D novelty and operational efficiency, striving for adaptive coordination rather than absolute balance (Venugopal et al., 2019; Liao et al., 2020). Without timely, coordinated seizing, even well-sensed opportunities dissipate; the more robust a firm's ambidextrous ability to grasp opportunities and put capabilities into practice, the greater a firm's potential to assert strategic agency in value creation and improve endogenous innovation outcomes (Medase & Barasa, 2019). In this sense, seizing is a firm's agility to materialise self-reliant innovation from within.

Reconfiguring denotes a firm's adaptive capacity to continually recombine, redeploy, and renew internal and relational assets to sustain innovation agility amidst systemic turbulence (Teece, 2007; Teece et al., 2016). In contrast to static restructuring, reconfiguring is a capability of an evolutionary process through the realignment of routines, structures, and resources (Wilden et al., 2016). Ambidexterity enables firms to exploit and bridge accessible resources with emergent innovation capability-building (Atzmon et al., 2022; Pundziene et al., 2022). Amidst the intensifying interplay of geopolitical fragmentation, the capacity for purposeful reconfiguration increasingly constitutes the fulcrum of organisational adaptability and competitive renewal. This dynamic imperative compels firms not merely to withstand exogenous shocks but to orchestrate proactive, path-breaking transformation across structural, procedural, and relational domains. Whether through agile asset redeployment (Lovallo et al., 2020), boundary re-engineering to reduce value impedance (Pundziene et al., 2022), or the iterative renewal of innovation routines, reconfiguring facilitates firms to evolve from structural rigidity towards strategic plasticity—a prerequisite for adaptive, self-propelling endogenous innovation.

Empirical studies substantiate the centrality of dynamic capabilities to endogenous innovation. Firms with stronger resource reallocation capabilities transcend others by overcoming organisational inertia and directing financial and managerial attention towards emergent opportunities, especially when exogenous market signals are subtle (Teece et al., 2016; Wilden et al., 2016). This is particularly salient in strategic industries, such as digital health or green tech, where incumbent firms must balance legacy infrastructures with the urgency to internalise disruptive technologies (Lovallo et al., 2020, p. 1371). Moreover, evidenced by med-tech sectors, the deployment of dynamic capabilities implies successful innovation ambidexterity (Farzaneh et al., 2022; Pundziene et al., 2022). Regarding digitalised, knowledge-intensive industries, dynamic competition and broad-spectrum innovation—anchored in continuous resource recombination and entrepreneurial management—distinguish high performers from those reliant on static positions (Venugopal et al., 2019; Petit & Teece, 2021). The findings converge on that, in environments of fragmented global value chains and technological sovereignty, internal dynamic capabilities render firms to transcend the limitations of disrupted external linkages and renew their sources of advantage (Petricevic & Teece, 2019, pp. 1170–1172).

Briefly, the micro-organisational foundation of endogenous innovation is structured by firm dynamic capabilities. Against global fragmentation, restricted knowledge exchange, and regulatory bifurcation, the corporate capacity to internally orchestrate sensing, seizing, and reconfiguring processes constitutes the very definition of innovation from within. As Teece (2007) notes, dynamic capabilities construct competitive edge in regimes of change. In this sense, firm capabilities are not only the micro-foundations of firm growth but a strategic enabler of national and regional endogenous innovation systems writ large.

1.2 Macro-environment of Endogenous Innovation: Positive Externalities and Institutional Complementarity

Positive externalities constitute the macro environments through which states mobilise scientific, organisational, institutional, and policy resources to facilitate endogenous innovation. Functioning as structured configurations of policy frameworks, market incentives, and organisational routines, externalities catalyse knowledge production and diffusion within nationally bounded, institutionally complementary environments (Lundvall, 1992; Freeman, 1995; Furman et al., 2002). They are shaped by the coherence between industrial policy, research capacity, and firm absorptive capability, with regional variations reinforcing or constraining innovation systems (Cooke et al., 1997; Liu et al., 2020). Rather than static infrastructures, systems evolve through reflexive coordination among research institutions, private firms, and governmental agencies, forming recursive innovation circuits adaptive to technological and geopolitical shifts (Etzkowitz & Leydesdorff, 2000; Antonelli & Colombelli, 2015). Empirical evidence suggests that such systems perform best when institutional complementarities—across regulation, infrastructure, and incentives—coalesce to support strategic autonomy in innovation processes (Erzurumlu et al., 2022; Wang et al., 2012). Moreover, while open innovation is often framed as globally integrative, its effectiveness depends on national systems' capacity to govern cross-boundary knowledge flows without compromising internal capability formation (Wang et al., 2012; Oh et al., 2016). Thus, in the context of techno-nationalism and the emerging Tech Cold War, national systems serve not only as mechanisms of innovation enhancement but also as instruments of technological sovereignty, embedding endogenous innovation within the broader geostrategic fabric of the state (Liu & White, 2001; Edgerton, 2007; Luo, 2022a).

Building upon these systemic foundations, regional industrial clusters and inter-firm ecosystems serve as spatial and organisational conduits through which national innovation systems are materially enacted and locally embedded. Industrial clusters create an intertwined landscape of cooperation and competition among high-tech firms and regional industry peers, wherein institutional environments significantly enhance firm development. While intra-industry rivalry is inevitable, it does not necessarily translate into destructive competition. When firms benefit from robust institutional support, efficient upstream–downstream coordination, and shared infrastructural platforms, such rivalry often becomes a catalyst for lean operations and differentiated innovation strategies (Lii & Kuo, 2016; Juhász & Lengyel, 2018; Thatchenkery & Katila, 2021). Proximity—both geographical and sectoral—fosters trust and accelerates communication. Formal and informal exchanges among co-located firms contribute to knowledge spillovers and the consolidation of industry standards (Juhász & Lengyel, 2018). In partner selection, firms frequently consider cognitive and strategic proximity to establish efficient collaboration networks (Binz & Truffer, 2017; Marra et al., 2019). These decentralised interactions, when sufficiently dense, catalyse the formation of regional innovation platforms, which in turn enhance joint innovation among

SMEs within clusters (Leckel et al., 2020; Vlaisavljevic et al., 2020; Osarenkhoe & Fjellström, 2022). Such locally embedded models of open innovation provide valuable cues for policy intervention—local governments can facilitate collaborative capacity by institutionalising regional platforms and industrial alliances.

Moreover, cross-sectoral and inter-regional partnerships along the value chain extend the breadth and depth of innovation networks, particularly when leveraged through digital platforms (Kazantsev et al., 2023). Close, trust-based relationships with suppliers and clients expedite product iteration and process refinement, enhancing firms' responsiveness to market volatility. The deployment of digital tools further enables synchronised innovation across the supply chain, thereby elevating firms' performance and market returns (Li et al., 2023; Kazantsev et al., 2023; Chakraborty et al., 2024; Yu et al., 2024). In parallel, cross-sectoral collaboration and technological convergence remain pivotal sources of innovation, as firms integrate frontier knowledge from disparate fields to pursue boundary-spanning, transdisciplinary breakthroughs (Koo & Le, 2024). Leading technology firms often spearhead the construction of platform-based ecosystems, integrating upstream, downstream, and cross-industry actors into unified innovation architectures. These complex ecosystems necessitate strong dynamic capabilities to overcome value creation frictions and realise network effects (Pundziene et al., 2022; Chakraborty et al., 2024). By orchestrating horizontal collaboration and vertical integration across supply chains, such firms consolidate their influence across entire industrial value chains, gaining leverage in both standard-setting and market access (Teece, 2022).

The interaction between firms and their surrounding clusters, as well as embedded innovation ecosystems, plays a critical role in long-term innovation performance. Embedded within regional clusters, high-tech firms draw upon the support of finely specialised value chains, as well as external knowledge infrastructure such as universities, public research institutes, incubators, skilled labour, and auxiliary services. Incubators and science parks, through financial support and access to entrepreneurial networks, significantly accelerate the scale-up of start-ups (Mungila Hillemane et al., 2019). Technological capabilities within high-tech sectors tend to spatially agglomerate. In such contexts, anchor firms often function as focal nodes, stimulating innovation dynamism across the cluster through collaborations with surrounding SMEs and start-ups (Blakely & Hu, 2019; Ye et al., 2020; Tu et al., 2023). Facing growing barriers to international expansion, many Chinese high-tech firms have strategically shifted their growth focus towards domestic markets by integrating into and reinforcing local innovation ecosystems (Zhang et al., 2023).

Linkages with global value chains offer technology firms access to broader markets and global knowledge reservoirs. Many high-tech SMEs gain exposure to international best practices through supplier roles within MNC networks, using open innovation as a pathway to foreign market entry (Pananond, 2016; Guo & Zheng, 2021; Simba et al., 2023). However, climbing the value chains ladder and capturing higher value-added activities requires proprietary core technologies and agile strategic orientation (Ju & Gao, 2022; Capatina et al., 2024). For firms in emerging markets, balancing local market cultivation with GVC participation is especially salient. In an increasingly fragmented global innovation landscape (Luo, 2022b), firms must recalibrate their strategies between localisation and globalisation—reducing dependency on foreign technologies by reinforcing indigenous innovation and integrating domestic value chains to build adaptability and secure long-term competitiveness (Zhang et al., 2023).

Besides, agglomeration economies and their positive externalities form a crucial substrate for the development of leading technology firms. Firms embedded in geographically proximate high-tech parks or industrial clusters benefit from shared labour pools and specialist supplier networks, reducing search costs and improving matching efficiency (Marshall, 1890; Krugman, 1991; Binz & Truffer, 2017; Xu & Hu, 2024). A concentrated talent base facilitates recruitment, while shared physical infrastructure—transport, telecommunications, and R&D facilities—reduces operational overhead. Although not created for any single firm, these conditions generate public goods-like benefits for the entire cluster (Pinch & Sunley, 2016; Martinus & Sigler, 2017; Giuliano et al., 2019). Knowledge spillovers are particularly pronounced in such environments, as neighbouring firms' R&D activities diffuse through both formal collaborations and informal interactions, raising the collective knowledge base (Leckel et al., 2020). When nearby firms exhibit high absorptive capacity, such spillovers foster a collaborative learning ecosystem that buffers against innovation failure (Ngo & Thornton, 2022; Garcia et al., 2024; Mate-Sanchez-Val et al., 2025). The presence of anchor firms also yields additional spillover effects, with talent mobility and technology diffusion elevating regional entrepreneurial dynamism (Lew & Liu, 2016; Fotopoulos, 2022; Howell, 2020).

In summary, the evolutionary growth of tech firms is structured by their capacity to integrate internal dynamic capabilities with the positive externalities in the interplay between internal capability-building and external resource clustering. Effective participation in regional clusters, supply chain ecosystems, and global value networks requires firms to orchestrate both local and transnational linkages while sustaining organisational agility. Only by simultaneously cultivating internal capabilities and leveraging positive externalities—local, regional, national, and global—can firms achieve technological innovation, adaptive growth, and competitive advantage.

2. Endogenous Innovation: A Chain Process

This research, contextualised in the fragmentation and realignment of global innovation chains, aims to revisit the endogenous innovation of emerging-markets firms with a dynamic growth capabilities framework. From a bottom-up micro-firm perspective, our revisit articulates the firm capabilities developing throughout three stages—*technology exploration*, *market exploitation*, and *ecosystem orchestration*. Our framework argues that innovation is, rather than a linear result of internal R&D or exogenous imitation-based knowledge regeneration, ambidextrous dynamic capability-building among R&D, business scaling, and ecosystem embedding. The three stages and both stage-specific and co-evolving, interwoven capabilities manifest as the dynamics by which emerging-markets tech lead firms convert available resources into endogenous innovation and growth.

2.1 Technology Exploration

As the rate and direction of innovation are endogenously technology-pushed (Comin et al., 2025), the technology exploration capability is the primary engine of firm integration into multinational supply chains. When emerging-markets firms are increasingly excluded from the global mobility of advanced technologies by the Tech Cold War, explorative innovation stands out at the helm. Endogenous innovation originates in explorative innovation constituting a strategic imperative for functional transformation in value chains (Zhang et al., 2023). Those surmount technological bottlenecks imposed by global techno-economic powers accrue competitive advantages in tech rivalry, market competition, and supply chain centrality (Mousavi et al., 2018; Ju et al., 2020; Capatina et al., 2024). The transformation from component supplier (specialised modular provider) to core supply chain ecosystem orchestrator hinges on this exploratory capability first, which is thus reconceptualised as comprising three micro-firm endogenous innovation modalities—original, recombinant, and incremental—three dynamic capabilities of exploration complemented by exploitation.

Original innovation capability denotes a science-based, *zero-to-one* exploration that translates fundamental scientific research into marketable, knowledge-intensive inventions (Patel & Pavitt, 1997; Jansen et al., 2006). This capability targets technological fronts where firms engage in high-uncertainty-high-impact components that is often radical in nature, creating breakthrough tech products, categories, or markets (Dosi, 1982; Henderson & Clark, 1990; March, 1991; Govindarajan & Kopalle, 2006). Firms with salient original innovation ability not only pioneer novel products but also influence the structural reconfiguration of industrial ecosystem, embedding themselves as indispensable supply chain tech nodes (Weerawardena et al., 2017; Capatina et al., 2024). Hwang et al. (2021) demonstrate that emerging-markets lead firms often coalesce open innovation ecosystem, publicly releasing prototypes or standards that attract complementary innovators, followers, and imitators. This first-mover innovation is particularly critical in upgrading to higher value-added components, where early entrants can transform innovative research into supply chain centrality and non-substitutability. Original innovation illustrates the first modality by which nascent tech firms can effectively ascend through explorative R&D if given appropriate policy and market signals (Schot & Steinmueller, 2018; Hekkert et al., 2020; Leckel et al., 2020).

Recombinant innovation capability refers to the architectural integration of existing technologies across modular components, without compelling frontier invention (Henderson & Clark, 1990; Hargadon & Sutton, 1997). It is cross-sectoral knowledge recombination, whereby firms incorporate innovations from other domains to restructure the technological base. This capability is tied to absorptive capacity that enables firms to access, assimilate, and integrate heterogeneous knowledge resources (Cohen & Levinthal, 1990; Zahra & George, 2002; Teece, 2022). Particularly, it is central to *digital convergence* with platform-based and ecosystem-embedded innovation paradigms (Yoo et al., 2010). For instance, a pharmaceutical company can integrate AI and ICT to reconfigure R&D pipelines without directly innovating in those sectors (Koo & Le, 2024). Through dynamic digital convergence capabilities of technological opportunity seizing, cross-domain adaptive learning, and platform-enabled reconfiguration, firms repurpose advanced technologies into proprietary configurations to bypass the cost of R&D, accelerate endogenous innovation gains, and elevate their supply chain centrality (Luo, 2022b; Mao et al., 2024; Koo & Le, 2024). Succinctly, recombination is an exploration modality for emerging-market tech firms not endowed with foundational research power to recombine accessible knowledge into productive architectural advantages through learning-by-interacting (Atzmon et al., 2022; Mira Godinho & Simões, 2023).

Incremental innovation is a commercialisation-driven, engineering-intensive process of innovation in high-velocity product cycles. The increment capacity reflects a firm's capability of iterative product and process optimisations through design-for-manufacturing alignment (Dosi, 1982). Absent recourse to frontier R&D and recombination, this exploration refines feasible designs into scalable production responsively and resource-effectively (Hobday, 1995), closing the gap between technical feasibility and operational deliverability. Operational increment implies production efficiency and market responsivity (March, 1991; Gans & Stern, 2003; He & Wong, 2004; Teece, 2007). This process involves fast-cycle iteration and tactical execution for marginal time-to-market advantages via agile, preemptive commercialisation to capture emergent demands. As Lee et al. (2017) observe, firms can lead not by radical innovation, but as the first to commercialise optimised products. Crucially, the increment of product adaption and process refinement is a capability, though marginalised in traditional *lower-tier* innovation metrics, increasingly salient for industrious latecomer emerging-markets firms equipped with demand-centric manufacturing dexterity (Teece, 2022; Simba et al., 2023).

To evaluate the technology exploration capability of high-tech lead firms within fragmented value chains, a preliminary indicator framework aligned with original, recombinant, and incremental innovation is proposed (Teece, 2022; Godinho & Simões, 2023). 1) R&D intensity reflects a firm's commitment to original frontier knowledge creation. 2) Patent quality indicates the novelty and impact of R&D outcomes. 3) Core technology ownership signals control over proprietary, non-substitutable technologies critical for supply chain gatekeeping. 4) Talent density enables knowledge creation, absorption, and cross-domain integration. 5) Frontier technology incorporation captures a firm's involvement in transformative technologies it employs to recombine productive architectures and embed in next-generation industrial ecosystems. 6) Commercialisation cycle time measures the firm's efficiency in incremental engineering scaling. 7) Strategic project engagement suggests alignment with high-risk, policy-driven R&D missions.

Technology exploration is the first capability of endogenous innovation within our framework. Emerging-markets firms can endogenously generate and reconfigure innovation through original, recombinant, and incremental modalities to solve systemic bottlenecks, bypass techno-industrial exclusion, and embed itself in supply chains. Exploration is neither linear nor exogenous but a endogenous process dynamically activated through firm-specific capability-building. Facing a fragmented global innovation chains, the technology capacity to initiate growth from within is essential for functional upgrading and, next, market expansion.

2.2 Market Exploitation

Following the triple technology exploration, the capability of market exploitation reveals the critical inflection juncture at which innovation is converted into tangible market share. It is the capacity that firms mobilise organisational resources, scale operations, and embed technological assets into market competitive edge (Lee et al., 2017; Weerawardena et al., 2017; Liao et al., 2020). Situated as the second stage in the dynamic growth chain, it represents the transition to commercial leverage. Although technology exploration embeds firms into value chains as component suppliers, the transitioning to market power is nontrivial or contingent; while emerging-markets firms—particularly nascent high-tech ventures—may leverage technological capabilities to fuel revenue growth or cross-border entry, it never automatically ensures long-term prosperity (Hult et al., 2019; Venugopal et al., 2019; Ju et al., 2020). Market exploitation marks the beginning of influence—where firms attempt to navigate distribution architectures and redefine value capture pathways (Ju et al., 2020; Tu et al., 2023). Yet, moving from regional strongholds to global markets demands a mastery of market power in supply chain integration, whereby firms use market dominance to induce reverse upgrading from upstream suppliers and downstream purchasers (Weerawardena et al., 2017; Mousavi et al., 2018). Therefore, market exploitation is not merely a business objective but about growing up towards market impact.

The concept of ambidextrous dynamic capabilities constructs an analytical frame to interpret this stage—a firm's ability to manage the tension between exploration and exploitation. As Dodourova and Zhao (2023) argue, firms that focus solely on exploratory innovation without consolidation risk dispersion, while those that overexploit existing resources without reinvention stagnate. Market exploitation thus also demands dynamic reconfiguration—aligning product-market fit, adapting business models, and shaping new geographic and industrial linkages. Agile firms leap from experimental technologies to commercial platforms, from R&D to customer impact, from being mere participants in supply chains to influencing their trajectories. *Gazelle company* exemplifies this leapfrogging of exploitation; such high-growth firms, typically emerging from tech-intensive early stages, effectively scale after crossing the liability of newness, often registering successful revenue growth or IPO. Their growth is rooted in velocity and versatility of commercialisation—the dynamic interface between technology and market by adaptively seizing niches and scaling fast—represents the bellwethers of regional innovation and absorptive capacity (Ju et al., 2020; Liao et al., 2020). *Gazelles* thriving on ambidexterity, convert R&D momentum into demand capture and simultaneously reinvesting feedback into exploratory refinement.

Similar to technology exploration, market exploitation is also a dynamic process and capabilities of ambidexterity evolving with institutional environments and the market structures of competition. It is a sequentially coupled growth capability interlocked with exploratory innovation, whereby firms continuously readapt commercialisation, market penetration, and value chain positioning strategies in response to technology-induced market disruption and demand heterogeneity (Teece, 2007; Gereffi & Lee, 2014; Ju et al., 2020). Innovation is demand-pulled (Comin et al., 2025); even firms with the public visibility of novel technological breakthroughs might fail to translate technology into sustained market power, particularly in business environments fragmented by heterogenous demands (Jenson et al., 2016; Ju et al., 2020; Tu et al., 2023). Effective exploitation entails cross-functional integration—R&D, marketing, and operational capabilities (Venugopal et al., 2019; Kwon et al., 2022). Therefore, exploitation co-evolves with exploration through customer-informed iterations, real-time market sensing, and strategic reconfiguration dynamically to penetrate, structure, and close the gap between technological upgrading and market power.

To assess market exploitation capability, a multidimensional framework aligned with value realisation is essential. 1) Revenue growth reflects immediate commercial signal of innovation performance. 2) Market scope, including geographic and sectoral diversity, indicates breadth of customer reach. 3) Global penetration velocity, measured by speed and scale of international market entry, signals strategic depth. 4) Market responsivity captures real-time adjustment to customer demands and competitor actions. 5) Customer conversion and retention rates reveal efficiency in user acquisition and long-term engagement. 6) Brand power and localisation capabilities assess cultural adaptability and regulatory legitimacy across regions. 7) Platformisation capacity—such as closed-loop ecosystems through modular offerings, and subscription models—reflects a firm's ability to control demand interfaces and embed value delivery. This exploitation enables a shift from transaction-driven sales to relationship-based strategic integration.

Market exploitation transforms firms from emergent modular provider into innovation-enabled market entrants with market power. At this stage, technology becomes embedded in business routines, customer behaviours, and value networks. It marks the transitioning of emerging-markets firms from novelty to necessity. However, exploitation itself is insufficient for competitive advantage and commercial leverage. Evolving from market entrants to system architects, firms must engage in system orchestration—integrating techno-industrial rules, standards, and networks of production and innovation. Firms that fail to institutionalise their influence may remain brilliant but brittle; those that succeed define the infrastructural and relational orchestration to innovate not just products, but markets and the way markets work.

2.3 Ecosystem Orchestration

As firms achieve the positional advantage of strategic centrality beyond initial technological breakthroughs and early-stage market expansion, the capability to orchestrate an ecosystem signifies long-term competitive advantage. Concerning today's fractured global value chains, competitive advantage shifts away from isolated R&D or market power and towards the dynamic capacity to reconfigure any available resources and coordinate upstream and downstream value networks (Petricevic & Teece, 2019). Ecosystem orchestration thus denotes a firm's ability to embed itself in not only market reach but the extensive institutional and technological architectures—to be a rule-maker rather than a rule-taker in its domain.

This strategic shift towards ecosystem-thinking is propelled by the recognition that no single actor can internalise all the knowledge, capabilities, or resources necessary for adaptive endogenous innovation. During the global techno-nationalist fragmentation—global innovation chains are splintering—self-reliant national or regional innovation ecosystem is a buffer against shocks (Luo, 2022a; Zhang et al., 2023). Cultivating organisational adaptability against such shocks entails continuously reallocating resources, integrating iteratively novel technologies, and reconfiguring collaborative networks (Shi, 2024). Hence, orchestration is governing a reciprocally enhancing *innovation ecology* whereby lead firms reconstruct standards, update rules, and distributes opportunities across the ecosystem (Petit & Teece, 2021; Cuervo-Cazurra & Pananond, 2023).

Operationally, ecosystem orchestration involves two primary and complementary means (Mousavi et al., 2018; Marra et al., 2019). One is *vertical integration*—externalising control over critical upstream and downstream segments of value chains to secure essential materials, technologies, and market channels. The other is *platformisation*—reconfiguring products or services into a platform that extensive partnership and customership networks can build on, positioning the actor-with-platform as a central ecosystem orchestrator. For example, a *unicorn* startup is typically small in size yet wields out-sized influence of technological and market power on its industrial ecosystem and cross-sectoral alliances. The two means are mutually inclusive: vertical control reinforces platform authority, while platform scaling lowers the costs and risks of coordinating diverse collaborators. Further, orchestration necessitates aligning diverse innovation resources and institutional supports. Mobilising collaborations with universities, government agencies, suppliers, and even competitors to share material and intellectual resources, co-develop novel technologies, shape regulatory standards, and diffuse innovation risks across the networks.

Specifically, when sensed opportunities of globalisation narrow, firms may reach inwards for regional innovation clusters and domestic value-chain partnerships as alternative arenas for growth (Li et al., 2024). Chinese high-tech firms shifted from reliance on global supply chains to developing domestically anchored innovation ecosystems to achieve technological self-reliance (Zhang et al., 2023). This inward process of seizing leverages regionally clustered resources and capabilities, enhancing knowledge spillovers and incubating innovation platforms with localised roots but global ambitions (Juhász & Lengyel, 2018; Mungila Hillemane et al., 2019; Ngo & Thornton, 2022; Xu & Hu, 2024). Meanwhile, close alignment with public research institutes, government agencies, and industrial alliances embeds a firm's functional upgrading in a supportive milieu that amplifies endogenous innovation capacity (Etzkowitz & Leydesdorff, 2000; Kafouros et al., 2015; Chakraborty et al., 2024).

Moreover, industrial digital transformation, with the rise of digital platforms and data-driven operations, implies that one can orchestrate innovation activities across distances and sectors with unprecedented efficacy. Digital tools strengthen inter-organisational integration to boost platform-based innovation collaborations and remain agile amidst technological and market changes (Pundziene et al., 2022; Kazantsev et al., 2023). For another, digital *institutional infrastructures*—e.g. common data standards and market regulations—enable firms to adapt, coordinate, and scale more flexibly in turbulent environments (Luo, 2022b; Teece, 2022). Practically, firms with robust digital orchestration capabilities often outperform their peers in time-to-market, customer integration, and multi-industrial expansion (Li et al., 2023; Mao et al., 2024; Jie et al., 2025).

The evaluation of ecosystem orchestration centres on the orchestrator role. 1) Platform integrator is reflected in the number of ecosystem participants, the scale of collaboration networks, and the power in technical or market standard-setting. 2) Supply chain strategic position captures one's structural power of control over crucial technological and material resources. 3) Resource configuration gauges whether one melds disparate assets and knowledge across industries or regions into coordinated innovation. Another is 4) the intensity of ones collaborative linkages—joint projects with universities or public labs—alongside 5) the degree of cross-sectoral high-tech fusion. The indicators portrays a lead firm's dual influence over technology and market outcomes, together with the means it overcomes the lack of resources and capabilities (Cuervo-Cazurra & Pananond, 2023).

In short, ecosystem orchestration epitomises the institutional embedding of endogenous innovation, thereby transmogrifying the firm from a market rule-taker into a systemic architect whose dynamic capabilities are inextricably enmeshed within polycentric co-evolving ecology and platform-mediated collaborations. Through resource agglomeration, capability recombination, and ambidextrous governance, such embedding confers not merely sustained competitive advantage but strategic adaptive change—insulating the firm against exogenous techno-nationalist global value chain ruptures.

2.4 Endogenous Innovation Chain

Emerging-markets tech lead firms are increasingly defined not merely by their tech or market strength, but by their dynamic growth capabilities to construct, govern, and consolidate an endogenous innovation ecosystem. These firms—at the vanguard of technological, organisational, and supply chain integration—leverage innovation to transform their supply chain positioning from component suppliers to ecosystem orchestrators, eventually assuming agenda-setting roles in rule formation and standards negotiation (Paliwal, 2010; Gereffi, 2015; Li et al., 2021; Wong & Lee, 2021). It suggests a staged logic for nurturing such leadership: firms often begin as technological explorer, progress into sustained innovation and business model development, and ultimately evolve into supply chain lead firm by coordinating horizontal and vertical integration and platform-based ecosystem governance (Mousavi et al., 2018; Marra et al., 2019; Rakas & Hain, 2019; Lovallo et al., 2020; Petit & Teece, 2021; Zhang et al., 2023).

From the vantage point of global value chain theory and dynamic capability logic, the growth trajectory of high-tech lead firms can be mapped onto a three-stage capability chain: technology exploration, market exploitation, and ecosystem orchestration. These phases correspond respectively to the transitions from pre-leadership, to emergent leadership, and finally to institutionalised leadership (Mousavi et al., 2018; Mungila Hillemane et al., 2019). This structured framework clarifies the shifting challenges and capability demands facing firms at different stages, and provides a robust basis for differentiated policy support (Venugopal et al., 2019; Hwang et al., 2021; Atzmon et al., 2022; Teece, 2022; Simba et al., 2023; Koo & Le, 2024).

First, technology exploration defines whether a firm can enter global production networks through unique technological offerings (Mousavi et al., 2018; Ju et al., 2020; Capatina et al., 2024). It encompasses original innovation (from scientific research to frontier products), recombinant innovation (integrating digital technologies such as AI into legacy industries), and incremental engineering innovation (rapid commercialisation and iteration) (Weerawardena et al., 2017; Lee et al., 2017; Luo, 2022b; Koo & Le, 2024; Jie et al., 2025). For resource-constrained firms, lean development models and rapid prototyping provide essential strategies to manage uncertainty and accelerate market fit (Silva et al., 2021). Moreover, early strategic influence on technical standards enables firms to shift from merely innovating to reshaping sectoral rules (Pundziene et al., 2022; Teece, 2022).

In the second stage, market exploitation becomes the lever by which innovation is converted into sustained commercial success. Key capabilities include product-market alignment, business model innovation, and regional/sectoral expansion (Lee et al., 2017; Weerawardena et al., 2017; Liao et al., 2020). Here, digitalisation plays an amplifying role: not only in enhancing customer acquisition and local market responsiveness, but also in enabling firms to construct closed-loop user ecosystems (Luo, 2022b; Yu et al., 2024). The transition from technology provider to supply chain orchestrator hinges on this capacity to synchronise internal innovation with external commercial dynamics (Hult et al., 2019; Venugopal et al., 2019; Ju et al., 2020; Kwon et al., 2022; Simba et al., 2023).

For the third one, ecosystem orchestration becomes the defining capability for supply chain dominance and structural influence. This entails upstream control over key components, institutional coordination across technical and policy domains, and platform-enabled resource integration (Mousavi et al., 2018; Lovallo et al., 2020; Leckel et al., 2020; Petit & Teece, 2021; Zhang et al., 2023; Yu et al., 2024). Regional embeddedness matters as Weller and Rainnie (2023) caution that technological upgrading alone does not ensure sustained value chain engagement; firms must anchor themselves in regionally governed knowledge ecosystems. Standardisation capacity also becomes critical, enabling firms to shape global rules while curating domestic innovation environments (Gereffi & Lee, 2014; Gereffi, 2015; Kapoor & Teece, 2021).

The above triple-staged capability chain—technology exploration, market exploitation, and ecosystem orchestration—offers a differentiated framework to grasp the evolution of tech lead firms. For ‘pre-lead’ firms, policy can prioritise scientific grants, pilot subsidies, and early-stage market access. For ‘emerging leaders’, support should be placed on scaling channels, digital transition, and market penetration. And for ‘institutionalised post-lead’ ones, policy must back global standard-setting, international mergers, and regional ecosystem governance. Only by calibrating policy to the rhythm of capability-building can emerging economies facilitate the next-generation innovation-driven, globally embedded lead firms.

3. How to Cultivate Endogenous Innovation?

Building on the dynamic capabilities framework articulated before, it turns to the challenge at the heart of contemporary innovation strategy. How can states and firms cultivate endogenous innovation against techno-nationalist fragmentation? The growth capabilities—technology exploration, market exploitation, and ecosystem orchestration—outlines the essential stages through which tech firms ascend from capability formation to systemic integration. Yet, realising it demands the construction of robust, adaptive ecosystems, purposeful institutional support, and the strategic harnessing of both local and transnational resources. Hence, several emerging pathways—spanning policy, infrastructure, regional absorptive capacity, and organisational agency—is unravelled.

3.1 Innovation Ecosystem

Techno-nationalism, or *neo-techno-nationalism*, is an emerging policy orientation by which states contest in the supremacy over innovations in technology amidst the Tech Cold War (Godinho & Simões, 2023; Tung et al., 2023). Techno-nationalism spotlights the centrality of technology superiority to national security and autonomy (Edgerton, 2007; Tung et al., 2023;

Zhang et al., 2023), characterised by state-directed initiatives. Neo-techno-nationalism, albeit recognising international interdependency, continues to concentrate state power on steering self-reliant techno-industrial innovation, blending nationalist objectives with reconstructing global economic and trade rules (Shim & Shin, 2016). It is also why endogenous innovation is not autarky but inclusive of global knowledge transfer and its contextualised regeneration (Awate et al., 2015; Kim et al., 2020; Zhao et al., 2022; Zhang et al., 2023; Li et al., 2025). A gauntlet thrown down to the US-dominated global economic order is *Made in China 2025* initiative that earmarked several high-tech industries for technology sovereignty in advanced value chains (Starrs & Germann, 2021). In response, incumbent powers deployed protective measures to restrict Chinese tech M&A and knowledge accession, while emerging economies mobilised to fortify domestic R&D capabilities (Awate et al., 2015; Li et al., 2024; Shi, 2024). This policy reveals the geopolitics intertwined with international business and national innovation systems (Moura et al., 2025).

A salient strategic initiative is on constructing a neo-techno-nationalist innovation ecosystem, essentially a national innovation system to facilitate endogenous innovation. As the rise of new techno-nationalism imposed exclusive competition against *open innovation* on global value chains (Edgerton, 2007; Luo, 2022a), the endogenous innovation ecosystem emerges as a state's policy to attain technology superiority. Based on this policy orientation, a neo-techno-nationalist innovation ecosystem is definable as 'a state-orchestrated and globally intertwined system wherein a state aligns techno-industrial innovation with national security and power expansion adaptive to evolving geopolitics' (Shim & Shin, 2016; Petricevic & Teece, 2019; Kim et al., 2020; Zhao et al., 2022; Shi, 2024). To be specific,

1) the state in this ecosystem aims at self-reliant endogenous innovation and to construct the system whereby it acts;
2) the state can mobilise domestic and transnational actor-networks to convert (de/re)globalisation into a source of competitive advantage, involving firms (e.g. Huawei, Xiaomi), governments, and non-human agents (AIs);
3) the neo-techno-nationalist innovation ecosystem can in turn fortify national techno-economic security and leverage both protective and cooperative means to raise the state's power of technology sovereignty.

Kim et al. (2020) articulate that China, through *state empowerment*, augmented the global connection of its communications industry by orchestrating synergistic government-industry collaboration in international 5G standards-setting. In addition to this standardisation, Anand et al. (2021) argue that the state is a *supportive embedded* agent within the innovation ecosystem, provisioning industrial policy, intellectual property protection, market access, and other institutional support for adaptive endogenous innovation. Godinho and Simões (2023, pp. 3–5) exemplify that China's ascendancy in critical techno-industrial domains was also buttressed by substantial state-directed initiatives and its national innovation system via direct financing and projecting on frontier sectors. The case studies evince a shift from a micro-firm perspective towards a co-evolutionary model of dynamic interactions among governments, firms, and the neo-techno-nationalist innovation ecosystem (Kafouros et al., 2015; Anand et al., 2021, p. 548), which is an organised system that emerging economies can emulate (Parente et al., 2021).

Neo-techno-nationalist endogenous innovation accelerates catching-up yet may risk isolating the domestic ecosystem from global knowledge integration. Although China is globally ahead in patent filings—processors, semiconductors, and communications—its innovation ecosystem remain peripheral in international R&D influence networks (Godinho & Simões, 2023). Yet, endogenous innovation never rejects globalisation but recalibrates it by re-embedding external knowledge within adaptive local systems, vice versa (Kafouros et al., 2015; Zhao et al., 2022). Zhang et al. (2023) highlight that international exposure—education and outbound R&D—lays a groundwork for innovation capabilities. Crucially, a resilient endogenous innovation ecosystem is achievable through strategic knowledge transfer in regionalised collaboration rather than US-centric globalisation. Regarding our economic multipolarisation, sustainable innovation systems can thrive by eclectically weaving innovation resources into locally adaptive architectures (Shi, 2024).

Methodologically, recent studies include quantitative patent data evaluations (Shim & Shin, 2016; Huang & Li, 2019; Godinho & Simões, 2023), firm- and industry-level case studies (Awate et al., 2015; Kim et al., 2020; Shi, 2024) , and systematic reviews (Zhao et al., 2022; Moura et al., 2025). Despite diverse approaches, a shared conceptual thread is *institutional context*. Endogenous innovation relies on supportive institutions, from robust intellectual property protection to mission-driven national projection (Hekkert et al., 2020; Anand et al., 2021; Godinho & Simões, 2023; Zhang et al., 2023). There is also a discernible consensus that geopolitical rivalry is a double-edged sword—it creates external pressure that stimulates endogenous innovation but also risks fragmenting the global innovation commons (Starrs & Germann, 2021; Tung et al., 2023). Li et al. (2024), with the balance-of-power theory to techno-nationalism, suggest that US containment policy considerably curtailed Chinese tech overseas M&A; in response, Chinese firms and the state doubled down on endogenous innovation, an internal balancing historically mirrored when external resources are restricted.

Concisely, the neo-techno-nationalist innovation ecosystem features strong state initiatives, targeted resource mobilisation, and selective international knowledge cooperation. It is a co-evolutionary ecosystem where firms innovate within institutional contexts open or not open to geopolitical dynamics. Balancing domestic R&D capabilities with global knowledge integration is vital for an adaptive endogenous innovation ecosystem.

3.2 Mega-science Infrastructure

Mega-science infrastructure is a tangible empowerment to facilitate endogenous innovation, which spills over original research capabilities into local knowledge production. It anchors endogenous innovation with scientific breakthroughs, especially by latecomer economies seeking to close the gap with techno-economic powers (X. Yang et al., 2024, pp. 1–2).

Empirical evidence substantiates the spillovers of original research capabilities. Zhang et al. (2024) summarise that a mega-science infrastructure, the Spallation Neutron Source at Guangdong, attracted and supported a cohort of biotech, pharmaceutical, and advanced manufacturing firms culminating in a cluster of nearly 500 entities with an output value of 24 billion RMB in 2022. The substantial scale of industrial agglomeration, the value-added, and enhanced original R&D capabilities collectively underscore the significant spillovers that research infrastructures exert on the endogenous innovation of high-tech firms (H. Yang et al., 2024). For another, X. Yang et al. (2024) estimate that the Shanghai Synchrotron Radiation Facility increased the share of Chinese publications in multiple domains by 12.4 percentage points, whilst its effect on international co-authored publications was negligible. This reveals that mega-science infrastructure is effective in amplifying domestic original research capabilities rather than international partnerships. Despite that, the enhanced research outputs driven by the unique experimental facilities can enable local-based scientists to originally advance in cutting-edge technologies that were previously unattainable (pp. 5–6). Besides, the UK, leveraging its synchrotron light source and spallation neutron source facilities, cultivated industrial clusters in nuclear, aerospace, and biotech sectors which generated substantial employment and innovation (Zhang et al., 2024)—credited with producing spillovers that benefit the broader economy from training highly skilled talents to spin-off high-tech innovations as new ventures (H. Yang et al., 2024). Therefore, X. Yang et al. (2024) claim that this infrastructure are transformative in enabling a state to progress from techno-economic dependency to self-reliant adaptability.

Mega-science infrastructure is not merely a cradle for high-tech firms but a catalyst for integrated innovation ecosystems and industrial clusters. By generating spillover effect, it drives the convergence of industry, innovation, and talent networks, embedding business actors within collaborative ecosystems that link universities, research institutes, firms, and governments (Kafouros et al., 2015; H. Yang et al., 2024; Zhang et al., 2024). Universities dig into frontier research; institutes focus on pilot testing; firms commercialise innovation outputs—together forming a seamless innovation network. Mega-science infrastructure also propels policy and industrial changes, as seen in the gas turbine test facility in Lianyungang, which advanced green technologies while guiding industrial green transformation (Zhang et al., 2024). Critically, the facilities institutionalise the innovation pipeline, bridging the *valley of death* between research and market. Government-directed roadmap and market-driven diffusion enable firms to undertake high-risk R&D within a vital ecosystem supported by preferential policies and intellectual property services. Thus, research infrastructure not only promotes technology breakthroughs but anchors a adaptive local innovation ecosystem.

However, the infrastructure-provisioning hinges on robust governance, sustained funding, and clear strategic alignment with the systematic goals of national innovation (X. Yang et al., 2024, p. 8). While mega-science infrastructure appears indispensable for building endogenous innovation capabilities, its long-term value relies on complementary policy toolkit that ensures research excellence is channelled into the local ecosystem.

3.3 Industrial Cluster

Knowledge transfer denotes the relocation and assimilation of valuable knowledge from an external source—a foreign subsidiary or host environment—into another. Emerging-market multinationals often establish overseas R&D units to acquire and internalise host-country tech knowledge and enhance home innovation (Li et al., 2025). *Absorptive capacity* delineates a firm's ability to identify, acquire, assimilate, transform, and exploit external knowledge for advantage (Cohen & Levinthal 1990; Zahra & George, 2002; Zhao et al., 2022). This capacity mediates the extent to which firms benefit from knowledge exchange of intentional transfer or incidental spillovers (Anand et al., 2021). Under neo-techno-nationalist ruptures of global innovation chains, cross-border R&D and external technology integration are precarious. Hence, endogenous innovation calls for intra-national knowledge circulation since robust regional absorptive capacities constitute the domestic knowledge transfer to yield adaptive innovation outcomes against geopolitical duress.

Although absorptive capacity was theorised in a context of cross-border knowledge transfer, a similar logic is salient with(in) national boundaries. China and other emerging economies exhibit intra-national inter-regional heterogeneity in resources, institutions, and capabilities, rendering disparate domestic knowledge transfer analogous to international diffusion. Contrary to assuming a homogeneous national system, sub-national institutional variations significantly mediate the efficacy of inter-organisational resource integration and innovation (Li et al., 2024). As at the international scale, resource integration necessitates the connectivity between knowledge sources and recipients, regions with stringent intellectual property enforcement and globally integrated research units yield superior innovation outcomes based on robust resource exchange networks (Kafouros et al., 2015). Conversely, geographic and institutional fragmentation may limit inter-regional knowledge diffusion and hinder endogenous innovation. Since techno-nationalism constrains global technology entanglement, the ability to mobilise domestic *boundary spanners* stands out (Liu & Meyer, 2020). Elite research institutions can translate tacit knowledge and bridge peripheral firms into usable forms. However, without absorptive infrastructures, similar collaborations may backfire. In essence, the principles of global knowledge transfer—recognition, connection, supportive institution—are determinative in governing inter-regional knowledge diffusion and sustaining endogenous innovation against techno-geopolitical decoupling.

Inter-regional absorptive capacity—the ability of regional innovation ecosystem to internalise and regenerate knowledge—critically delineates national innovation outcomes. Its efficacy hinges on the institutional architecture and knowledge exchange across regions (Anand et al., 2021). This capacity emerges from the synergistic interplay of firm capabilities, inter-organisational linkages, regional networks, and institutional supports. Firms must cultivate robust internal R&D and organisational learning while concurrently leveraging extrinsic knowledge networks (Liao & Li, 2018; Venugopal et al., 2019; Hwang et al., 2021; Teece, 2022). When innovation resources and capabilities are spatially asymmetric, domestic knowledge transfer is a key to endogenous innovation, highlighting multiple interrelated elements behind this internal-external synergy (Lassen & Laugen, 2017; Liao et al., 2020; Simba et al., 2023; Xu & Hu, 2024):

1) synergy between a firm's internal capabilities and adaptive organisational changes,

2) benign knowledge spillovers among co-located firms in the same industry,

3) linkage with supply chain partnerships across industries and regions,

4) integration into regional clusters, innovation ecosystems, then global value chains, and geopolitical dynamics, and

5) shared infrastructures and knowledge spillovers arising from industrial agglomeration.

These elements suggest absorptive capacity is not merely an internal firm attribute but embedded within the regional ecosystem. A firm's ability to utilise external knowledge hinges on its organisational capacity; even in open innovation, collaboration efficacy hinges on technical and managerial capabilities (Venugopal et al., 2019; Ju et al., 2020; Knight & Cuganesan, 2020; Atzmon et al., 2022; Kwon et al., 2022). Co-location with innovative firms and geographic proximity create trust and enable frequent formal and informal interactions, enhancing knowledge sharing and collective learning (Lii & Kuo, 2016; Juhász & Lengyel, 2018; Thatchenkery & Katila, 2021). Besides, digitised cross-regional, cross-industrial linkagesaugment innovation responsivity (Leckel et al., 2020; Kazantsev et al., 2023). Institutional architectures, including governments and industry associations as a platform, facilitate network connectivity and knowledge co-creation (Marra et al., 2019; Leckel et al., 2020; Vlaisavljevic et al., 2020; Osarenkhoe & Fjellström, 2022). Therefore, domestic knowledge mechanisms mirror global transfer, anchored in multi-scalar capabilities and institutions.

Domestic knowledge transfer and spillovers augment endogenous innovation beyond the confines of formalised in-house R&D. In regional industrial clusters, firms reap benefits from proximate R&D diffusion via both formal mechanisms and informal relational channels (Pinch & Sunley, 2016; Martinus & Sigler, 2017; Giuliano et al., 2019; Leckel et al., 2020). Yet, the efficacy of such spillovers is contingent on absorptive capacity; in high-capacity environments, a fertile knowledge-sharing milieu emerges and mitigates firm-level risks. A critical vector in this process is talent mobility of tacit know-how embedded in human capital. Anchor firms catalyse regional innovation by diffusing expertise through personnel and know-how circulation, thereby enhancing the entrepreneurial and technological dynamism of the innovation ecosystem (Howell, 2020; Fotopoulos, 2022; Ngo & Thornton, 2022; Garcia et al., 2024; Mate-Sanchez-Val et al., 2025). Given that such tacit knowledge evades codification, interpersonal transfer—via labour mobility and informal networks, and supplier-client interactions—is dominant. Historically, the innovation emerging-markets firms relied on absorbing and imitating external knowledge. China’s catching-up trajectory relied on sourcing, digesting, reverse engineering, and absorptive learning of foreign technologies (Li et al., 2025). Confronted with neo-techno-nationalist fragmentation, the decentralised, tacit-based exchange is indispensable. Provided sufficient absorptive infrastructures, peripheral regions can progressively converge with core hubs, enabling a more equitable, domestically anchored endogenous innovation ecosystem despite exogenous decoupling.

Endogenous innovation roots in absorptive capacities at multiple levels: firm-level, regional, and institutional. At the firm level, strong internal absorptive capacity grounded in R&D investment, skilled human capital, and organisational learning enables a firm to identify and exploit external innovation resources (Li & Yu, 2022). Regionally, the broader absorptive capacity of a regional innovation ecosystem—reflected in a skilled labour pooling, elite research institutions, and active inter-firm networks—facilitates knowledge (re)generation within and between regions. Local absorptive capacity mediates the impact of outside knowledge via inward foreign investment or inter-regional spillovers on regional innovation outcomes (Garcia et al., 2024). Meanwhile, institutions leveraged by government influence knowledge transfer through formal policymaking: for instance, requiring multinationals to set joint ventures or share technology with local partners can facilitate knowledge diffusion, whereas weak intellectual property enforcement might encourage informal spillovers or deter investments (Anand et al., 2021). These institutional determinants directly affect the ability of local firms and incoming actors to recognise, assimilate, and apply knowledge. According to China's evidence, a firm's growth and innovative outcomes arise from a synergy between internal capabilities and external structures, enabled by local, regional, national, and even global linkages (Lew & Liu, 2016; Fotopoulos, 2022; Howell, 2020). Endogenous innovation entails simultaneously strengthening a firm's own knowledge base and take advantage of multidimensional support structures—from regional industrial clusters and talent pools to national innovation systems and global networks. The multidimensional co-evolution of absorptive capacity creates a fertile environment wherein knowledge is effectively converted into novel products, processes, and long-term competitive advantages.

Absorptive capacity anchors the means knowledge circulation translates into innovation, globally or regionally. Inter-regional knowledge transfer mirrors global dynamics of firm capabilities, institutional endowments, and network linkages that jointly determine innovation outcomes. Strengthening these capacities is thus essential for building adaptive, inclusive national innovation systems of endogenous innovation and regional catching-up.

3.4 Mission-oriented Organisational Culture

Though the neo-techno-nationalist innovation ecosystem and state-directed initiatives set the stage, the micro-firm capabilities are equally a key to endogenous innovation confronted with geopolitical pressure. Corporate political capability and organisational culture are interrelated at a micro-firm level. Such internal capabilities and resources enable firms to navigate external turbulence such as government sanctions or market blockades and to continue innovating despite pressure. A corporate with adaptive political capabilities and solid pro-innovation cultures are better positioned to sustain and accelerate innovation in the Tech Cold War context.

Corporate political capability, according to Moura et al. (2025), refers to a firm's strategic competence to navigate, manage, and leverage the political environment to advance its objectives. It embodies an endogenous organisational capacity rooted in cumulative organisational learning, experience, adaptation, and deliberate investment to buffer or convert geopolitical challenges into resources and opportunities. Rather than merely reacting to policy shifts, politically capable firms align their strategies with evolving national initiatives to secure support and legitimacy; it is institutionally embedded in firm-government interactions dynamically over time and heterogeneously difficulty to imitate (Li & Lu, 2020; Moura et al., 2025). Ultimately, corporate political capability functions as a critical intangible asset that strengthens a firm's ability to cultivate and sustain endogenous innovation.

Corporate political capability enables endogenous innovation, particularly in the geopolitical frictions and policy volatility contexts. This perspective shifts the narrative from firms being passive victims of geopolitics to active agents that innovate around or through political constraints. Politically astute firms, equipped with robust governmental networks and regulatory literacy, are better positioned to secure advantageous treatment, mobilise policy instruments, and sustain their innovation trajectories amid external shocks. This firm-generated capacity determines the means external knowledge or resources from foreign and domestic governments are harnessed (Li et al., 2008; Krammer & Jiménez, 2019; Zhang & Guo, 2019; Zhao et al., 2024). For instance, during the US-Sina techno-trade tensions, firms adept in navigating political know-how and networks obtained exemptions or additional domestic support, allowing them to maintain or even expand R&D investment to replace restricted foreign technologies (Hu et al., 2024). Conversely, firms without political acumen were caught flat-footed by abrupt market access loss that impaired their innovation. Empirical studies from emerging markets corroborate that political connections facilitate corporate access to procurement contracts, subsidies, and state-backed knowledge production initiatives (Claessens et al., 2007; Goldman et al., 2013). These networks enable firms to tap into government grants and render regulations to favour domestic R&D (Zhang & Guo, 2019). However, while political capability offers significant leverage, over-dependence on individual connections introduces fragility when institutionalised agile political networks-building and continuous learning are absent. Thus, political capability should not be misconstrued as a static asset but rather as a dynamic organisational competence that, when effectively embedded, equips firms to transform political adversity into fertile ground for innovation.

Second is organisational culture, specifically that oriented towards adaptive innovation and a sense of mission. In an era of techno-nationalism, many Chinese tech firms cultivate a culture of patriotism and collective purpose to motivate innovation. An illustrative case centres on a Chinese high-tech firm navigating the Tech Cold War. The firm's indigenous innovation was driven by cultural attributes: a strong patriotic ethos, an elitist commitment to excellence against global competitors, and endurance in hardship (Zhang et al., 2023). This organisational culture became a form of informal institution that steered employee behaviour when formal institutions like open markets and supply chains faltered. Such vision-driven and mobilisation-oriented culture can drive innovation. It channels intrinsic motivation and aligns individual goals with the firm's innovation objectives, which is especially crucial when extrinsic incentives are dampened by external restrictions.

Intangible elements of vision, value, and identity significantly affect a firm's innovative capacity (Rindova & Martins, 2017). Against external threat in the context of geopolitical tension, techno-nationalism-infused vision statements or a resurgence of corporate mission often go beyond profit (Shim & Shin, 2016; Hekkert et al., 2020; Starrs & Germann, 2021). While concerns arise around echo chambers or insularity, the research highlights its positive role in galvanising effort and morale. Zhang et al. (2023) conclude that organisational culture works alongside R&D investment, technical skills, and effective resource use in explaining innovation outcomes under duress. For example, a firm with a risk-averse or complacent culture may fail even a large R&D budget, whereas a firm with a mission-driven culture may achieve breakthroughs with limited resources. This contrast suggests that adversity can strengthen innovation when filtered through a unifying culture.

In short, corporate political capability and pro-innovation organisational culture jointly constitute a firm's internal adaptive capacity to navigate external adversity and sustain endogenous innovation. A patriotic or mission-driven culture can galvanise employee support for political engagement, while effective political manoeuvring can, in turn, reinforce organisational cohesion. Crucially, this perspective broadens the conventional understanding of endogenous innovation beyond technical competency to encompass socially embedded assets—relationships, institutional know-how, and collective ethos—that reside within a firm. To cultivate genuine endogenous innovation, firms must therefore invest in not only R&D but building political acumen and nurturing adaptive, learning-oriented cultures. By weaving political capability and cultural vitality into endogenous innovation, firms and states can transform internal strengths into durable engines of growth amid geopolitical volatility.

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